MIL-C-39029/13A(AS)
2 August 1976
SUPERSEDING
MIL-C-39029/13(NAVY)
15 October 1973

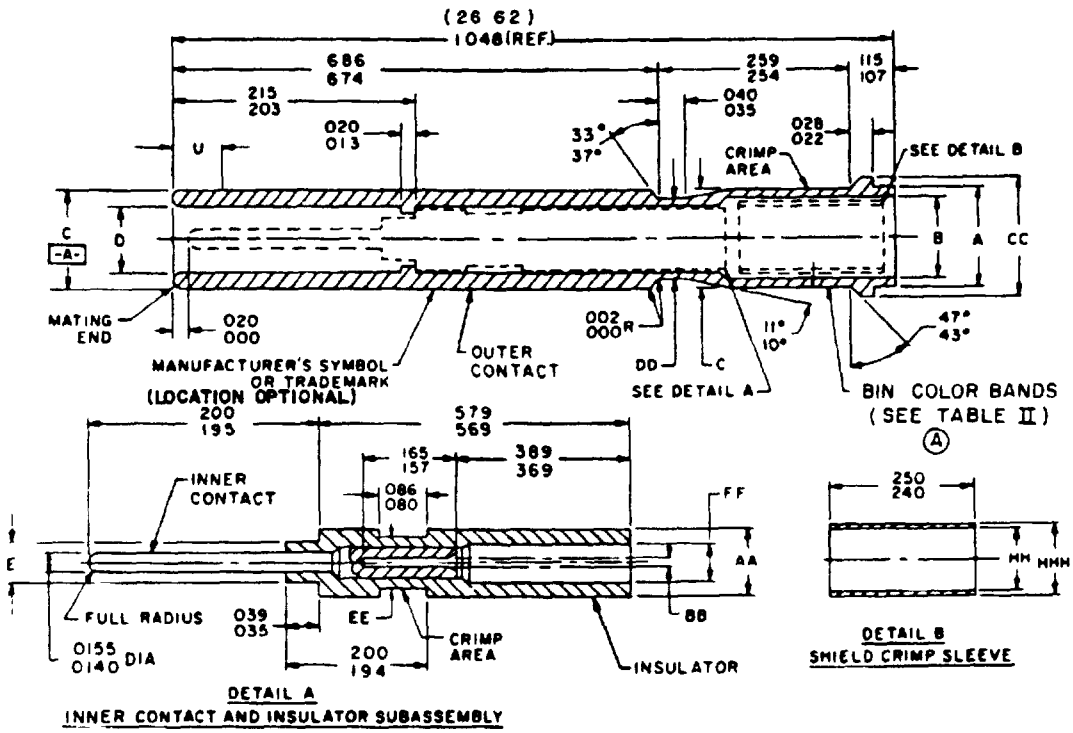
MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTOR,
SOCKET, CRIMP REMOVABLE, SHIELDED, (FOR
MIL-C-81511 SERIES 1 CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the contacts described herein shall consist of this document and the latest issue of Specification MIL-C-39029

INACTIVE FOR NEW DESIGN



NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses for overall length and diameter only.
4. Dimensions shown apply after plating.

FIGURE 1 Connector contact

FSC 5935

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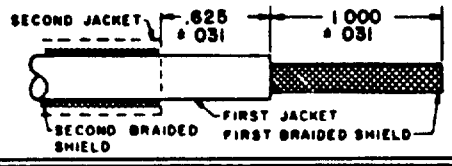
TABLE I. Dimensions.

BIN code	A DIA	B DIA	C DIA	D DIA	E MAX DIA	AA DIA	BB MIN DIA	CC DIA	DD DIA	EE DIA	FF MIN DIA	HH MIN DIA	HHH MAX DIA	U
152	.103	0910	.113	.0750	.042	.081	.022	.133 (3.38)	.099	.068	.037	.058	.087	.042
	.101	0885	.110	.0725		.078		.130 (3.30)	.096	.062				.018
153	.162	1430	.161	.1210	070	.128	.022	.190 (4.83)	.145	.076	.067	.086	.130	.042
	.159	1405	.158	.1175		.126		.187 (4.75)	.142	.070				.018
154	.162	1430	.161	.1210	.070	.128	.022	.190 (4.83)	.145	.076	.093	.110	.130	.042
	.159	.1405	.158	.1175		.126		.187 (4.75)	.142	.070				.018
155	.162	1430	.161	.1210	.070	.128	.022	.190 (4.83)	.145	.076	.108	.128	.142	.042
	.159	.1405	.158	1175		.126		.187 (4.75)	.142	.070				.018
156	.162	1430	.161	1210	.070	.128	.034	.190 (4.83)	.145	.089	.098	.128	.142	.042
	.159	1405	.158	1175		.126		.187 (4.75)	.142	.085				.018
157	.162	1430	.161	1210	.070	.128	.034	.190 (4.83)	.145	.089	.098	.110	.130	.042
	.159	.1405	.158	.1175		.126		.187 (4 75)	.142	.085				.018

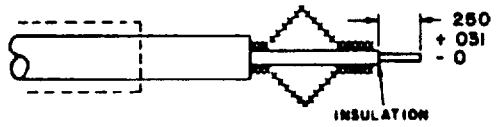
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STEP 1
1A STRIP FIRST CABLE JACKET BACK 1 INCH
1B IN THE CASES WHERE TRIAXIAL CABLES ARE USED THE SECOND JACKET AND BRAIDED SHIELD SHALL BE STRIPPED BACK AN ADDITIONAL 5 8 INCH

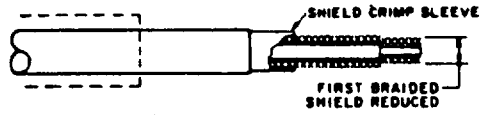
CAUTION DO NOT CUT INTO BRAIDED SHIELD



STEP 2
2A SLIDE FIRST BRAIDED SHIELD BACK AND TRIM INSULATION 1 4 INCH

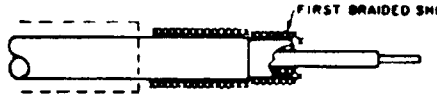


2B SLIDE FIRST BRAIDED SHIELD FORWARD AND ROLL BETWEEN FINGERS TO REDUCE ITS DIAMETER
2C. SLIDE SHIELD CRIMP SLEEVE OVER BRAIDED SHIELD UNTIL IT RESTS AGAINST THE FIRST JACKET



FOR CABLES USING M 39029/13-02,03 OR 06 CONTACTS	FOR CABLES USING M 39029/13-01,04 OR 05 CONTACTS
$2D_1$ NO CRIMPING NECESSARY	$2D_2$ CRIMP SHIELD CRIMP SLEEVE USING CAVITY B IN CRIMPING TOOL

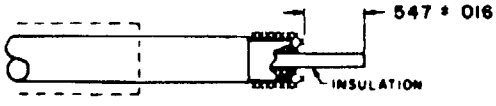
2E COMB OUT THE BRAIDED SHIELD AND FOLD IT BACK OVER THE SHIELD CRIMP SLEEVE



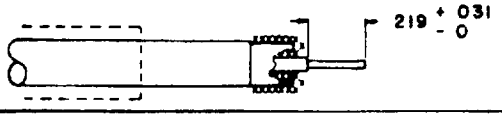
2F TRIM EXCESS BRAIDED SHIELD EXTENDING BEYOND THE SHIELD CRIMP SLEEVE



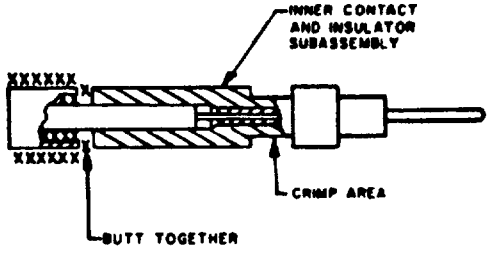
STEP 3
3A CUT INNER CONDUCTOR AND INSULATION



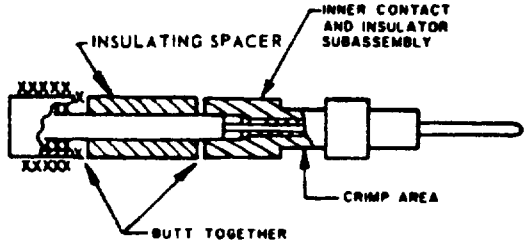
3B. STRIP INSULATION FROM INNER CONDUCTOR



STEP 4 CONTACTS M39029/13-01, -02, -03, -05 AND -06
4A RESTORE INNER CONDUCTOR TO ORIGINAL LAY AND INSTALL INTO INNER CONTACT AND INSULATOR SUBASSEMBLY BUTT CABLE AND CONTACT AS SHOWN
4B CRIMP INNER CONTACT THRU INSULATOR WITH CRIMPING TOOL AND POSITIONER LISTED IN TABLE III.



STEP 4 CONTACT M39029/13-04
4A. INSTALL SPACER OVER INSULATION AGAINST BRAIDED SHIELD
4B. RESTORE INNER CONDUCTOR TO ORIGINAL LAY AND INSTALL INTO INNER CONTACT AND INSULATOR SUBASSEMBLY BUTT CABLE, SPACER AND CONTACT AS SHOWN
4C CRIMP INNER CONTACT THRU INSULATOR WITH CRIMPING TOOL AND POSITIONER LISTED IN TABLE III



STEP 5
5A INSERT CRIMPED SUBASSEMBLY INTO OUTER CONTACT UNTIL SOLID RESISTANCE IS FELT AND OUTER CONTACT COVERS FIRST BRAIDED SHIELD
5B CRIMP OUTER CONTACT USING CAVITY A IN CRIMPING TOOL.

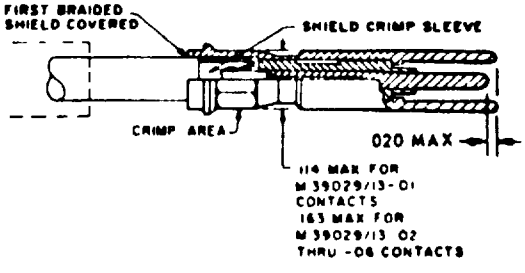


FIGURE 2 Assembly procedure

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2 Aug 1976

TABLE II Design characteristics

BIN code	Color bands			Contact cavity size	Cable accommodated	Type	Class
	1st	2nd	3rd				
152	Brown	Green	Red	16	RG-178A/U RG-196A/U	D	B
153	Brown	Green	Orange	12	RG-179B/U RG-187A/U RG-188A/U		
154	Brown	Green	Yellow	12	9530D5117 (RAYCHEM) 1/		
155	Brown	Green	Green	12	RG-180B/U RG-195A/U 293-3922 (MICRODOT) 1/		
156	Brown	Green	Blue	12	250-4070 (MICRODOT) 1/		
157	Brown	Green	Violet	12	5022F5111 (RAYCHEM) 1/		

1/ or equivalent

REQUIREMENTS

Dimensions, design characteristics, and configuration See figure 1 and tables I and II

Tools See table III

Mating contact MIL-C-39029/6

Manufacturer's recommended assembly instruction to be shipped with unit package.

Assembly procedure See figure 2

Preparation of samples Contacts shall be wired as required using wire in accordance with table II (cables accommodated column)

Contact resistance Contact resistance at a load current of 1.0 ampere shall meet the requirements of table IV

Contact engagement and separation forces The contact separation and engagement forces shall meet the requirements of table IV.

Dielectric withstanding voltage 1,000 volts, ac, rms from sea level to 110 000 feet altitude

Tensile strength The tensile strength shall meet the requirements of table IV

QPL evaluating activity Naval Weapons Support Center, Code 3074, Crane, Indiana 47522

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TABLE III. Tools

BIN code	Inner contact		Shield crimp sleeve		Outer contact		Installing tool	Removal tool
	Basic crimping tool	Positioner	Basic crimping tool	Die	Basic crimping tool	Die		
152	M22520/2-01	M22520/2-26	M22520/5-01	M22520/5-02 Cavity B	M22520/5-01	M22520/5-02 Cavity A	MS3323-16	MS3344-16
153	M22520/2-01	M22520/2-25	M22520/5-01	M22520/5-04 Cavity A	M22520/5-01 M22520/10-01	M22520/5-04 Cavity A M22520/10-04 Cavity A	MS3323-12	MS3344-12
154	M22520/2-01	M22520/2-25	M22520/5-01	M22520/5-04 Cavity A	M22520/5-01 M22520/10-01	M22520/5-04 Cavity A M22520/10-04 Cavity A	MS3323-12	MS3344-12
155	M22520/2-01	M22520/2-25	M22520/5-01 M22520/10-01	M22520/5-04 Cavity B M22520/10-04 Cavity B	M22520/5-01 M22520/10-01	M22520/5-04 Cavity A M22520/10-04 Cavity A	MS3323-12	MS3344-12
156	M22520/2-01	M22520/2-25	M22520/5-01 M22520/10-01	M22520/5-04 Cavity B M22520/10-04 Cavity B	M22520/5-01 M22520/10-01	M22520/5-04 Cavity A M22520/10-04 Cavity A	MS3323-12	MS3344-12
157	M22520/2-01	M22520/2-25	M22520/5-01	M22520/5-04 Cavity A	M22520/10-01 M22520/10-01	M22520/5-04 Cavity A M22520/10-04 Cavity A	MS3323-12	MS3344-12

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TABLE IV Test limits

BIN code	Tensile strength (lbs) min		Contact engaging and separation forces (ounces)				Maximum contact resistance (millivolts)	
			Inner contact		Contact assy			
	Inner contact	Outer contact	Min	Max	Min	Max	Inner contact	Outer contact
152	6 0	15	0 5	7 0	2.0	20 0	8 0	4 0
153	6.0							2.0
154	16 0							
155	6 0							
156	32 0							
157	17 0							

Military part numbers See table V

TABLE V Military part number

Bin code	Military part number
152	M39029/13-01
153	M39029/13-02
154	M39029/13-03
155	M39029/13-04
156	M39029/13-05
157	M39029/13-06

Custodians
Navy - AS

Preparing activity
Navy - AS

Agent
DSA - ES

(Project 5935-N041-1)

Review activities
Navy - EC, SH, MC
DSA - ES

User activities
Navy - OS

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